



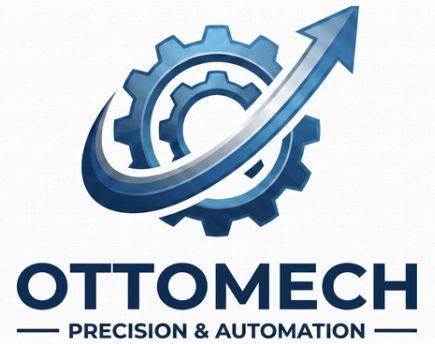
OTTOMECH
— PRECISION & AUTOMATION —

Your Mechanic. One Tap Away.

Track — Open Innovation | Team — 4 Members

Because breakdowns don't wait.

THE PROBLEM



Stranded. Stressed. Exploited.

32.5 lakh

Registered Vehicles in UP

Urban: 30–45 min
Rural: 60–120 min

Min Avg Breakdown Wait Range

0

Trusted Mechanic Platforms

No Discovery

No real-time way to find a nearby mechanic when stranded on the road.

No Trust

No ratings, no credentials, no accountability for roadside repairs.

No Transparency

Price negotiated under panic. Quality unknown until after the job.

UP-32 Lucknow is one of UP's largest vehicle hubs; and there's no single trusted mechanic platform in the city.

Supply Exists. Demand Exists. The Connection Layer Doesn't.

60+ multi-brand garages in Lucknow (Sulekha), 146 in Lalbagh alone (Justdial), 89 authorized service centers (ZigWheels); Many lack online booking, live availability, or dispatch capabilities. We're not creating supply. We're connecting it.

Drivers (Demand Side)

Lucknow commuters, highway travellers on NH27/NH30, tourists, gig economy riders. Pain: helpless when stranded. Need speed, trust, fair price.

Mechanics (Supply Side)

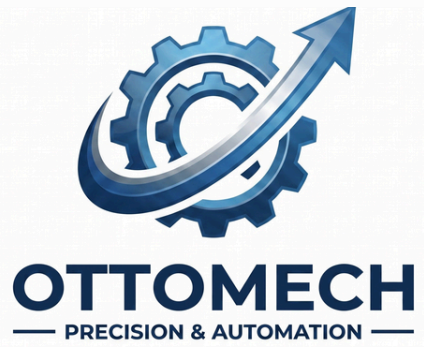
Hundreds of listed garages with no inbound job pipeline. No digital identity. No payment trail. Pain: invisible to the internet economy.

Market Signal

11.7% CAGR

DataIntel projects the Asia-Pacific mobile mechanic services market to grow at an 11.7% CAGR from 2026–2034, driven by rising vehicle ownership and increasing adoption of on-demand automotive services

HOW IT WORKS



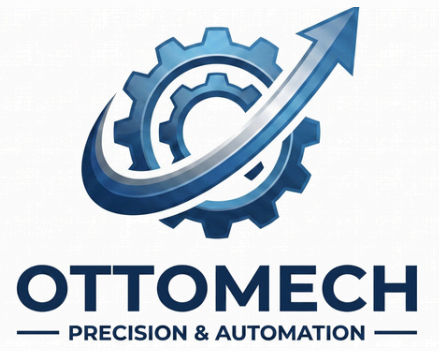
Four Steps. Under Five Minutes.

OttoAssist connects stranded drivers to nearby verified mechanics — no app install, works on any smartphone browser.



From first tap to mechanic on the way — under five minutes, on any device.

KEY FEATURES



Built for the Breakdown Moment

Issue Pre-Selector

Tyre / Battery / Engine / Overheating / Other.
Mechanic arrives prepared.

Upfront Pricing

Quote locked before job confirmation. No surprise bills.

One-Tap SOS

Broadcasts job to 3 nearest available mechanics simultaneously.

Damage Photo Upload

Driver sends photo at issue-input stage, reducing diagnostic time on arrival.

Digital Receipt + Warranty

PDF auto-generated post-job. Mechanic can flag warranty period. Driver sees it in-app.

In-App Chat

Direct message between driver and mechanic after match confirmation.

What Sets OttoAssist Apart



Mechanic Reliability Index (MRI)

Proprietary trust score based on on-time arrival, job completion rate, customer ratings, response speed, and warranty history.



Offline Mode

Cached mechanic data remains accessible in low-connectivity areas using Service Workers and IndexedDB. No internet needed to view mechanics or initiate a call.

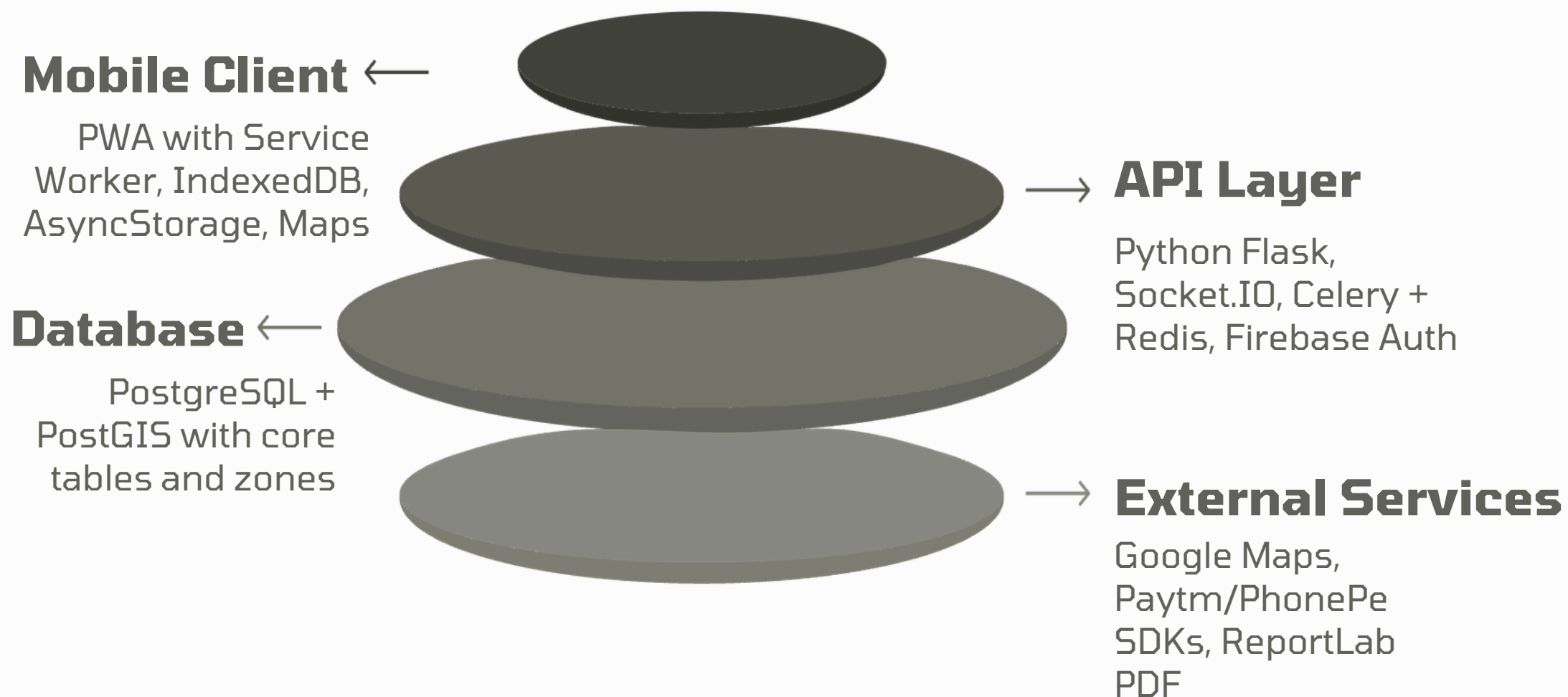


Digital Receipt + Warranty

Every completed job generates a digital receipt containing mechanic details, pricing, parts used, GPS location, and timestamp.

❏ Offline support ensures users can still access previously cached mechanic information when connectivity is poor.

Four-Layer PWA Architecture



Design Decisions

Offline resilience at the client layer via Service Workers + IndexedDB caching.

Real-time tracking via Socket.IO WebSocket connections for live mechanic pin movement.

Spatial discovery via PostGIS radius queries on PostgreSQL — "find nearest mechanic" is a geo problem, not a filter.



Real Data. Not Dummy Data.

Step 1 — Source: Pulled real Lucknow garage listings from Justdial (146), Sulekha (60), ZigWheels (89 centers).

Step 2 — Seed: Curated 20 verified garages across Gomti Nagar, Hazratganj, Alambagh, Indira Nagar, Transport Nagar, Chinhath, Lalbagh. Real names, addresses, coordinates into PostgreSQL.

Step 3 — Activate: 8 garages tagged "online" for demo. 22 as standby. PostGIS returns nearest 3–5 in real time.

IMPACT & BUSINESS MODEL

For Users

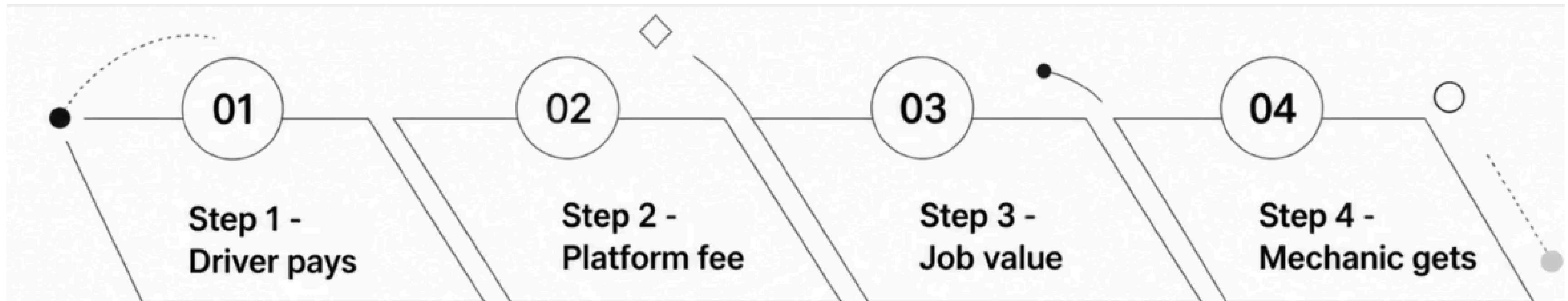
60–90 min wait → under 15 min. Upfront price. Live tracking. No exploitation.

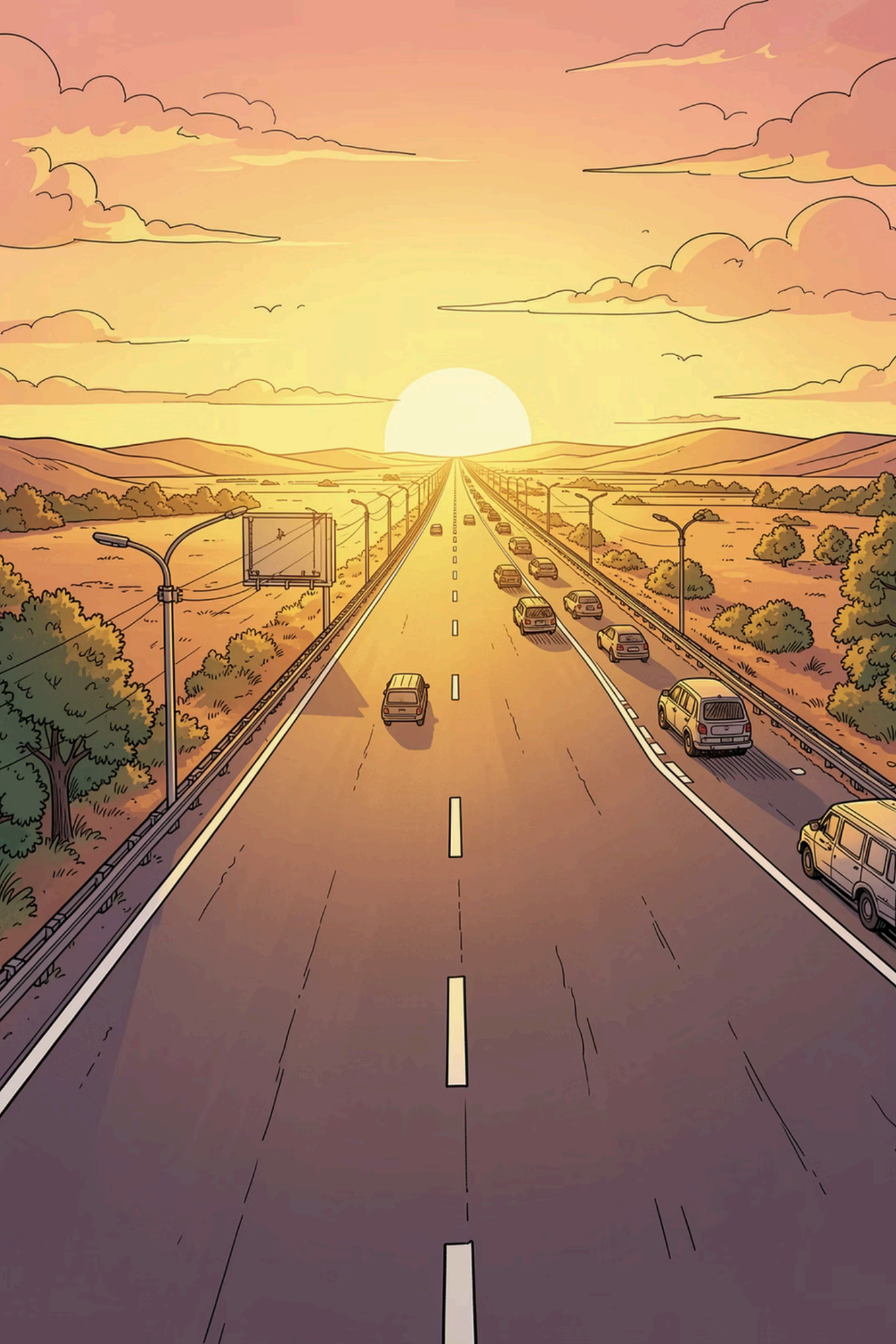
For Mechanics

Digital profile. Steady job pipeline. Instant UPI payout. Payment trail for creditworthiness.

For Lucknow

Quality floor raised by MRI. Road safety improved. Unorganized sector gets a data layer.





ROADMAP & CLOSING

Phase 1 — Now (Hackathon Prototype)

Core loop end-to-end. 30 real Lucknow garages seeded. Offline mode live. MRI V1 scoring. Payment sandbox.

Phase 2 — 6 Months (Lucknow Live)

Mechanic onboarding field ops across 8 zones. Voice job alerts (TTS). Warranty dashboard. NH27/NH30 highway corridors activated.

Phase 3 — 12 Months (UP Expansion)

Rollout to Kanpur, Agra, Varanasi, Prayagraj. Fleet/B2B API for logistics. Job queue batching. Earnings analytics. All 75 UP districts mapped.

Closing Statement

We didn't build for India. We built for the driver stranded on Faizabad Road at 11pm with no network.

OttoMech
works
when
nothing
else does.



Offline mode isn't a stretch goal. It's the point.